



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
REVIEW TEST
03-07-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

PAPER-1

Time : 3 Hours

Maximum Marks : 252

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **34** pages and that all the **22** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I(i) : (Maximum Marks : 24)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. An uncharged parallel plate capacitor having a dielectric of constant K is connected to a similar air cored parallel capacitor charged to a potential V . The two capacitors share charges and the common potential is V' . The dielectric constant K is :

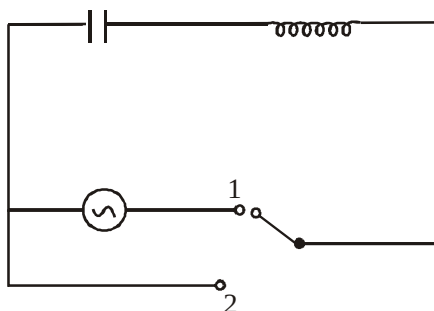
(A) $\frac{V' - V}{V' + V}$ (B) $\frac{V' - V}{V'}$ (C) $\frac{V' - V}{V}$ (D) $\frac{V - V'}{V'}$

2. The magnetic flux through a stationary loop with resistance R varies during interval of time T as $\phi = at(T - t)$. The heat generated during this time neglecting the inductance of loop will be :-

(A) $\frac{a^2 T^3}{3R}$ (B) $\frac{a^2 T^2}{3R}$ (C) $\frac{a^2 T}{3R}$ (D) $\frac{a^3 T^2}{3R}$

Space for Rough Work

3. Consider a series LC circuit connected with AC potential of $V_0 \sin(\omega t)$ as shown in figure. There is a two way switch initially connected to point 1. It may be shifted to position 2.



(A) Maximum charge on capacitor after switch is shifted to position 2 does not depend on time of shifting.

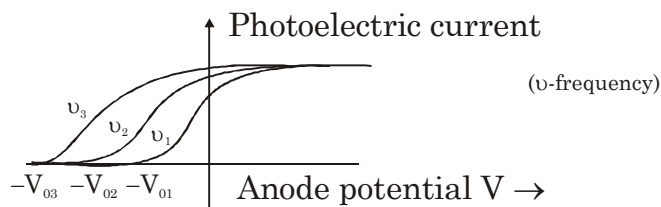
(B) Maximum current in inductor after shifting of switch is $V_0 \sqrt{\frac{C}{L}}$

(C) Time period of oscillation of current in circuit after shifting of switch is $\frac{2\pi}{\omega}$.

(D) Maximum charge on capacitor after shifting of switch is

$$Q_{\max} = \frac{V_0 C}{(1 - \omega^2 LC)} \left[\sin^2 \omega t + \omega^2 LC \cos^2 \omega t \right]^{1/2}$$

4. In photoelectric effect, the curve between photoelectric current and anode potential V (for different frequencies) is shown in figure, then



(A) $v_1 > v_2 > v_3$

(B) $v_1 < v_2 < v_3$

(C) $v_1 = v_2 = v_3$

(D) $v_1 > v_2 < v_3$

Space for Rough Work

5. In a meter bridge experiment the resistance of resistance box is $16\ \Omega$, which is inserted in right gap. The null point is obtained at 36 cm from the left end. The least count of meter scale is 1mm. The value of unknown resistance is

(A) $9 \pm \frac{5}{128}\ \Omega$

(B) $9 \pm \frac{5}{250}\ \Omega$

(C) $9 \pm \frac{5}{512}\ \Omega$

(D) $9 \pm \frac{1}{2560}\ \Omega$

6. Two identical bubbles stick with each other and are in equilibrium, their radius is R. The area of common interface is equal to

(A) $\frac{\pi R^2}{4}$

(B) $\frac{3\pi R^2}{4}$

(C) $\frac{2\pi R^2}{3}$

(D) None

Space for Rough Work

7. A uniform elastic string has length a_1 when tension T_1 and length a_2 when tension is T_2 . The amount of work done in stretching it from its natural length to a length $(a_1 + a_2)$ is :

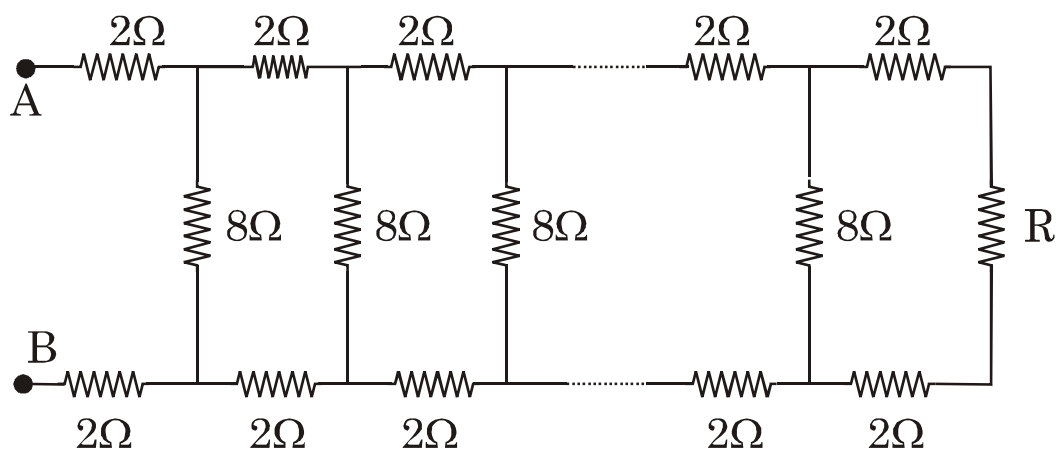
$$(A) \quad \frac{1}{2} \frac{(a_1 T_1 + a_2 T_2)^2}{(T_1 - T_2)(a_1 - a_2)}$$

$$(B) \quad \frac{1}{2} \frac{(a_1 T_1 - a_2 T_2)^2}{(T_1 - T_2)(a_1 - a_2)}$$

$$(C) \frac{1}{2} \frac{(T_1 + T_2)^2}{(T_1 - T_2)(a_1 - a_2)}$$

$$(D) \frac{1}{2} \frac{(T_1 + T_2)}{(T_1 - T_2)(a_1 - a_2)}$$

8. Consider the ladder network shown in figure. What should be the value of resistance R, so that effective resistance between A & B becomes independent of number of elements in the combination?



- (A) 2Ω (B) 4Ω (C) 8Ω (D) 16Ω

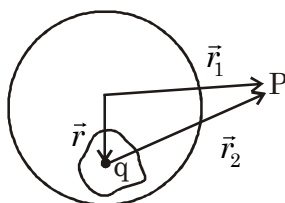
Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 32)

- This section contains **FOUR** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

Paragraph for Questions 9 and 10

An uncharged solid spherical conductor of radius R centered at origin has a cavity of some arbitrary shape is formed as shown in the figure. Somewhere within the cavity charge q lies as shown.



9. The field at P outside the sphere due to induced charge on the outer surface of sphere is

(A) $\frac{kq\vec{r}_1}{r_1^3}$ (B) $kq\left[\frac{\vec{r}_1}{r_2^3} + \frac{\vec{r}_2}{r_2^3}\right]$ (C) $kq\left[\frac{\vec{r}_1}{r^3} + \frac{\vec{r}_2}{r_2^3}\right]$ (D) $kq\left[\frac{\vec{r}}{r_1^3} - \frac{\vec{r}_2}{r_2^3}\right]$

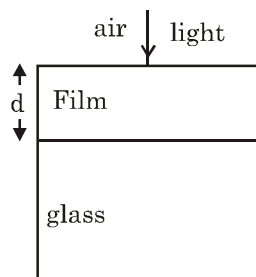
10. The potential at point P due to induced charges is :

(A) $kq\left[\frac{1}{r_1} - \frac{1}{r_2}\right]$ (B) $kq\left[\frac{1}{r_1} + \frac{1}{r_2}\right]$ (C) $kq\left[-\frac{1}{r_1} + \frac{1}{r_2}\right]$ (D) $\frac{kq}{r_1}$

Space for Rough Work

Paragraph for Questions 11 and 12

Light having wavelength ' λ ' is incident on thin film coated glass. Thickness of film is d . Refractive indices of film and glass is 1.2 and 1.5 respectively. Light is incident normally.



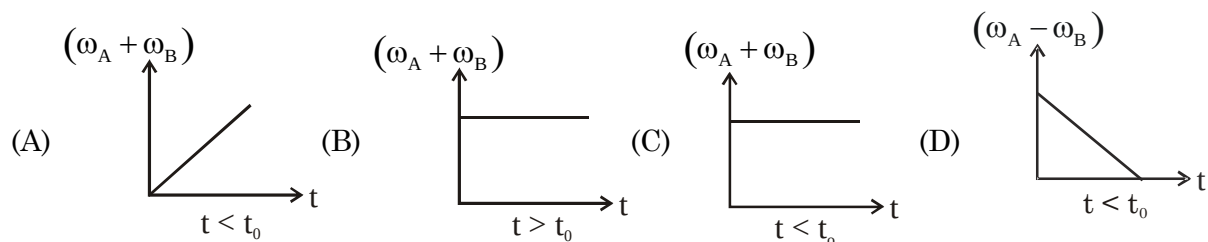
11. Minimum thickness of film to get destructive interference on reflection side is
- (A) $\frac{\lambda}{1.2}$ (B) $\frac{\lambda}{2.4}$ (C) $\frac{\lambda}{4.8}$ (D) $\frac{\lambda}{3.6}$
12. If intensity of incident light is I_0 and 20% light is reflected on each interface, then maximum intensity of light on reflection side will be (consider only two waves for interference).
- (A) $\frac{49}{125} I_0$ (B) $\frac{81}{125} I_0$ (C) $4I_0$ (D) $\frac{36}{125} I_0$

Space for Rough Work

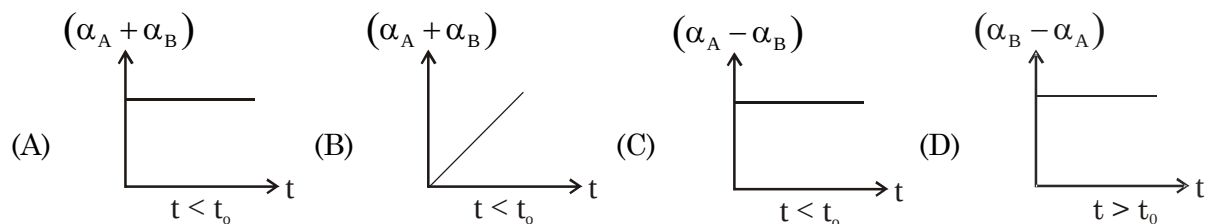
Paragraph for Questions 13 and 14

A uniform disc A is placed horizontally on a smooth horizontal surface. It is rotating with angular velocity ω_0 about a vertical axis passing through centre of mass. Now another identical disc B is placed gently and coaxially on disc A. Disc B is initially at rest and coefficient of friction between discs is μ . ω_A and ω_B are angular velocities of A and B at time t . Relative sliding stops at time t_0 . Take direction of ω_0 as positive.

13. Which of the following graphs is/are **CORRECT** ?



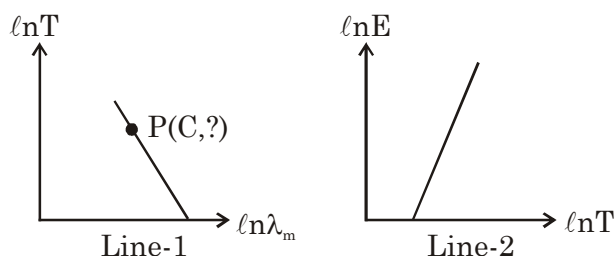
14. If α_A and α_B are angular accelerations of discs then which of the following graphs is/are **INCORRECT**?



Space for Rough Work

Paragraph for Questions 15 and 16

The variation of the $\ln T$ versus $\ln \lambda_m$ is given, where λ_m is the wavelength corresponding to which maximum number of photons are emitted and E is the energy emitted by the body per second and T is the temperature. The intercept made by the line l on the Y-axis has length A .



15. What is the value of Wein's displacement constant ?

- (A) e^A (B) $\frac{1}{e^A}$ (C) $\ln A$ (D) $\frac{1}{\ln A}$

16. A black body is emitting radiations at certain temperature. If the radiation emitted by the body is represented by point P on the line l , then value of ordinate of point P is

- (A) $C - A$ (B) $A - C$ (C) A (D) C

Space for Rough Work

SECTION-II : (Maximum Marks: 12)

- This section contains **FOUR** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

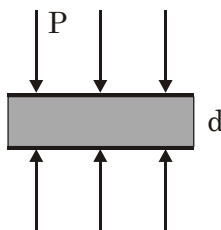
For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	●	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

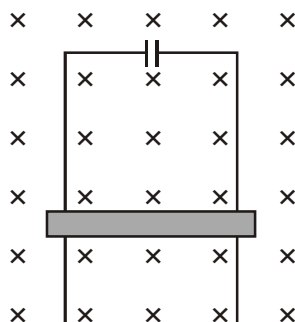
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If **ONLY** the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A capacitive pressure gauge is shown in figure. Two plates of area A are separated by a material with dielectric constant κ , thickness d, and Young's modulus Y. If a pressure increase ΔP is applied to the plates, the fractional change in capacitance is given by $\frac{\alpha \Delta P}{y}$. The value of $|\alpha|$ is



Space for Rough Work

2. A capacitor of capacitance $100 \mu\text{F}$ is connected to smooth railing and connector of mass 1 kg and length 1 m , whole system is in vertical plane and a magnetic field of strength 100 T is applied perpendicular to plane of circuit as shown. Connector is released under gravity. The time (in sec) after which dielectric of capacitor will burn out is $k \times 10^3$ then find the value of k . Breaking strength of dielectric is $5 \times 10^8 \text{ V/m}$. (d is the distance between plates of capacitor = 1 mm)



3. A hot-air balloon stays in equilibrium because hot air at atmospheric pressure is less dense than cooler air at the same pressure. If the volume of the balloon is 500.0 m^3 and the surrounding air is at 15.0°C , what should be temperature of the air in the balloon for it to lift a total load of 290 kg (in addition to the mass of the hot air) (in k) is [The density of air at 15.0°C and atmospheric pressure is 1.23 kg/m^3]
4. Gold, $^{198}_{79}\text{Au}$, undergoes β^- decay to an excited state of $^{198}_{80}\text{Hg}$. If the excited state decays by emission of a γ photon with energy 0.412 MeV , what is the maximum kinetic energy (in MeV) of the electron emitted in the decay? (The recoil energy of the $^{198}_{80}\text{Hg}$ nucleus can be ignored. The masses of the neutral atoms in their ground states are 197.968225 u for $^{198}_{49}\text{Au}$ and 197.966752 u for $^{198}_{80}\text{Hg}$.)

Space for Rough Work

SECTION-III : Integer Value Correct Type
No question will be asked in section III

SECTION-IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- The ORS contains a 4×5 matrix whose layout will be similar to the one shown below :

(A)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(B)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(C)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(D)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)

- For each entry in **column-I**, darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

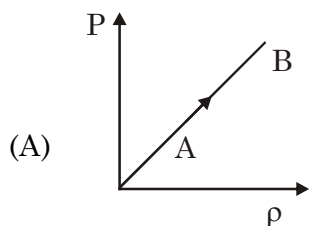
Full Marks : +2 If only the bubble(s) corresponding to all the correct match(es) is (are) darkened

Zero Marks : 0 In all other cases

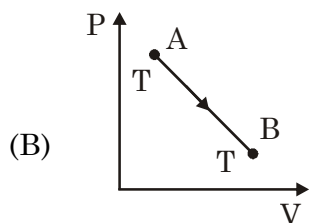
1. For a given gas. Match column - I with column II.

Column-I

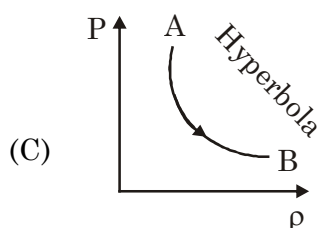
Column-II



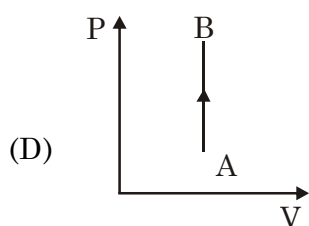
(P) Internal energy is constant



(Q) Work done is positive



(R) Work done is negative



(S) Internal energy continuously increases

(T) Heat is absorbed by the gas

Space for Rough Work

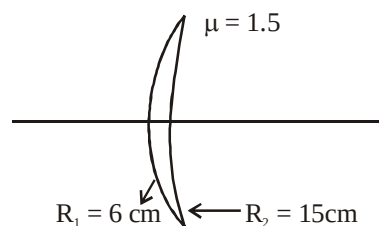
2. Column-I contain position of object which may be real or virtual on principal axis of optical system mention in Column-II. Match the correct options.

Column-I

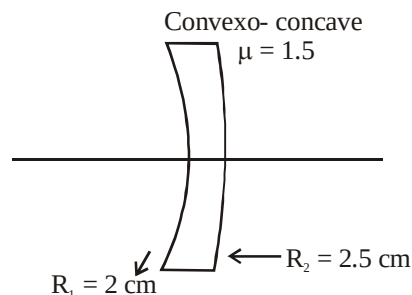
Column-II

Concavo - Convex

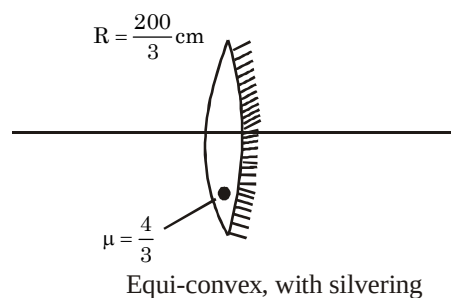
- (A) Object at $u = -20$ cm, image is at infinity (P)



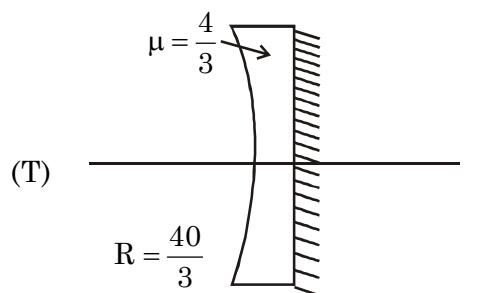
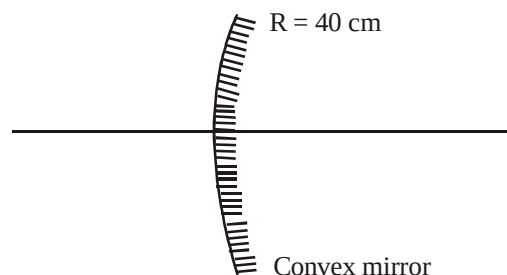
- (B) Object is at distance 40 cm from system, its image may be at 40 cm from system. (Q)



- (C) When object is at $u = -10$ cm from system its image is at 20 cm distance from system. (R)



- (D) System has +ve optical power (S)



Plane-concave, with silvering

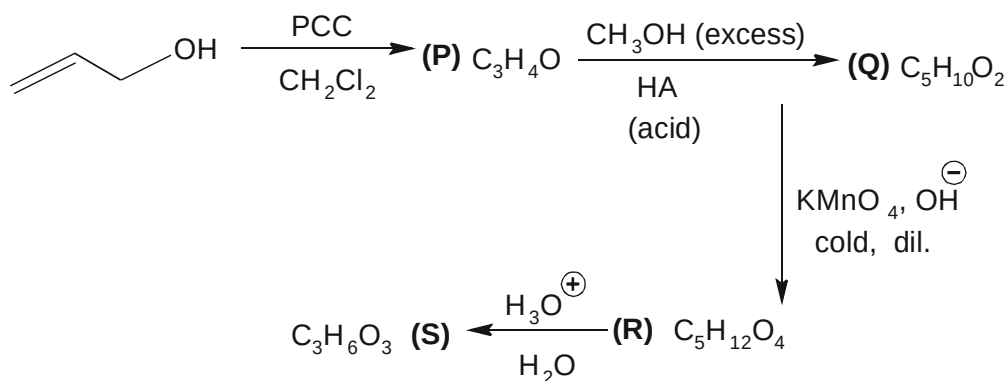
Space for Rough Work

PART-2 : CHEMISTRY

SECTION-I(i) : (Maximum Marks : 24)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
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1. A solution containing 1.23 g of $\text{Ca}(\text{NO}_3)_2$ in 10 g of water has a normal boiling point of 100.975°C . The degree of dissociation of $\text{Ca}(\text{NO}_3)_2$ will be : ($K_{b, \text{water}} = 0.52 \text{ k kg mol}^{-1}$)
 (Ca = 40; N = 14, O = 16)
 (A) 0.25 (B) 0.33 (C) 0.5 (D) 0.75
2. Reaction scheme :



Correct statement(s) about above reaction scheme is ?

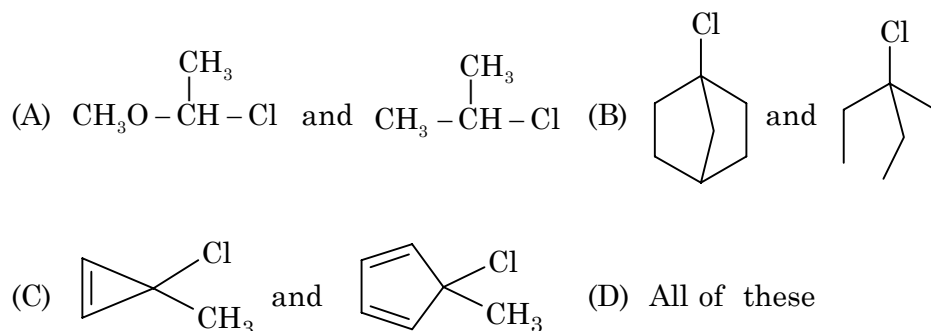
- Compound (Q) is an Acetal
- Compound (S) when reacted with LiAlH_4 followed by H_2O gives glycerol.
- Compound (P) is Acrolein.
- All of the above

Space for Rough Work

3. Which of the following is the correct order of second ionisation potential ?
- (A) $\text{Be} < \text{Li} < \text{B} < \text{C}$ (B) $\text{Mg} < \text{Al} < \text{Si} < \text{Na}$
 (C) $\text{C} < \text{N} < \text{F} < \text{O}$ (D) $\text{Si} < \text{P} < \text{S} < \text{Cl}$
4. The rate of a first order reaction is 0.4 M sec^{-1} at 10 minutes and 0.04 M sec^{-1} at 20 minutes after initiation. The half life of the reaction is about
- (A) 2 minutes (B) 3 minutes (C) 6.9 minutes (D) 3.3 minutes
5. Which statement is incorrect :
- (A) In ethylene glycol gauche form is more stable than antiform due to intramolecular H-bonding.
 (B) Antiform and gauche form have same torsional strain in butane about $\text{C}_2 - \text{C}_3$ bond.
 (C) $\text{X}-\text{CH}_2-\text{CH}_2-\text{X}$ [where X is halogen] having antiform more stable than gauche form about C-C bond, for all halogen.
 (D) For skew form dihedral angle is $0^\circ < \theta < 60^\circ$
6. Which of the following mixtures has predominantly "Keesom forces" between the molecules?
- (A) Benzene + Carbon tetrachloride (B) Nitric acid + Carbon tetrachloride
 (C) Sodium chloride + water (D) Nitric acid + chloroform

Space for Rough Work

7. Standard enthalpy of neutralisation of H_3PO_4 against KOH is -11 kcal/eq. then standard enthalpy of ionisation of H_3PO_4 in kcal / mole is (standard enthalpy of neutralisation of strong acid with strong base is -13.7 kcal/eq.)
- (A) 2.7 kcal / mole (B) 5.4 kcal / mole
 (C) 8.1 kcal / mole (D) 1.7 kcal / mole
8. Choose the pair in which compound II has more reaction rate for $\text{S}_\text{N}1$ mechanism as compare to compound I.



Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 32)

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- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
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Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

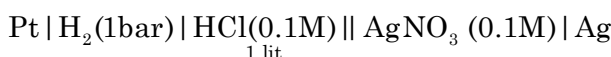
Paragraph for Questions 9 and 10

Galvanic cell is an arrangement where chemical energy is converted in electrical energy. Its

e.m.f. is given by Nernst equation. $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{2.303 RT}{nF} \log_{10} Q_{\text{pc}}^{\circ}$

If $E_{\text{cell}} > 0 \Rightarrow$ cell reaction is spontaneous ; $E_{\text{cell}} = 0 \Rightarrow$ equilibrium is established
 $E_{\text{cell}} < 0 \Rightarrow$ cell reaction is non-spontaneous

9. For given galvanic cell at 25°C



If 4 gm of NaOH is added into L.H.S. compartment, then change in e.m.f. of the cell is :

(Use : $\frac{2.303 RT}{F} = 0.06$)

- (A) 0.18 V (B) - 0.36 V (C) - 0.18 V (D) 0.36 V

10. For given galvanic cell



$$E_{\text{Cl}^-|\text{Hg}_2\text{Cl}_2|\text{Hg}|\text{Pt}}^{\circ} = 0.28\text{V} \quad (\text{Use : } \frac{2.303 RT}{F} = 0.06)$$

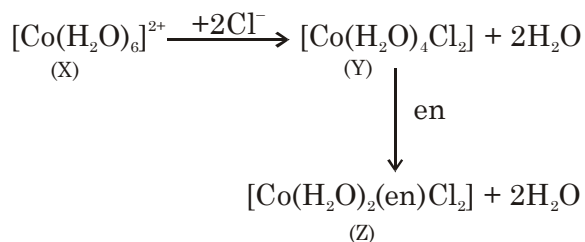
Select the correct options.

- (A) Hydrogen electrode is anode
- (B) Calomel electrode is cathode
- (C) On diluting HCl solution 10 times, e.m.f. of the cell remains unchanged
- (D) For HCl conc. 0.1 M and hydrogen gas pressure 10^{-4} bar, $E_{\text{cell}} = 0.28\text{ V}$

Space for Rough Work

Paragraph for Questions 11 and 12

Isomerism in co-ordination compound is also equally possible like isomerism in organic chemistry. Structural as well as stereoisomerism are possible for octahedral, square planar and tetrahedral complexes. One reaction sequence is given below :

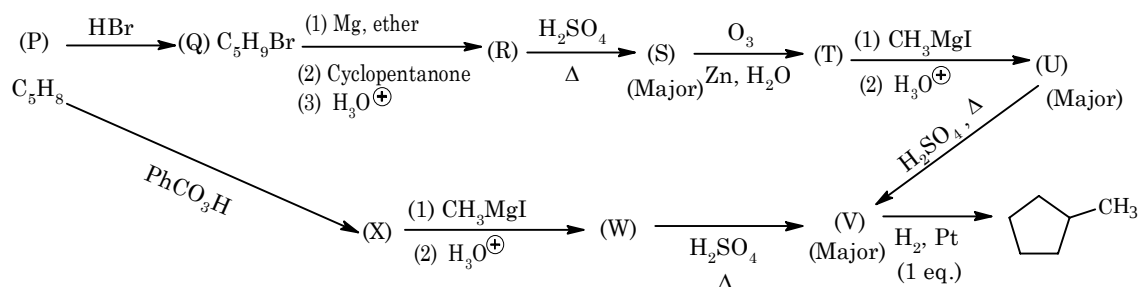


11. The number of $\widehat{\text{O}-\text{Co}-\text{O}}$ angles (90°) in X, Y(cis) and Z(cis w.r.t. to H_2O) variety are respectively (in idealised form)
- (A) 8, 5, 1 (B) 12, 6, 1 (C) 12, 5, 1 (D) 8, 6, 3
12. The optical activity can not be shown by :
- (A) X and cis variety of Y (B) cis variety of Y and trans variety of Z
- (C) All G.I. of Z (D) All G.I. of Y

Space for Rough Work

Paragraph for Questions 13 and 14

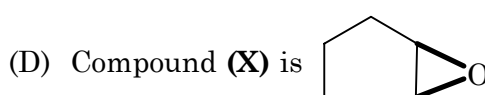
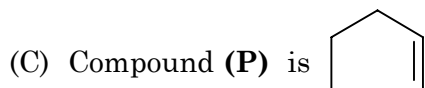
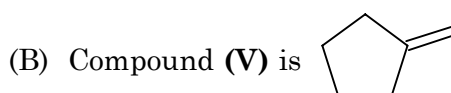
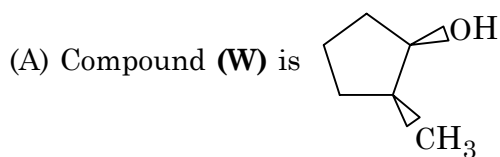
Reaction scheme :



13. Select the correct statement(s) for above reaction scheme.

- (A) Compound (R) and Compound (U) both are tertiary alcohols.
 (B) Compound (X) is chiral compound.
 (C) Compound (T) can show Tautomerism.
 (D) Compound (P) gives a same product with HBr and HBr / R_2O_2

14. Identify the correct match.



Space for Rough Work

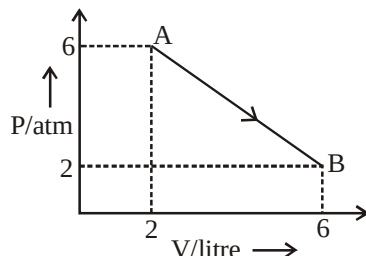
Paragraph for Questions 15 and 16

For a reversible process including only P-V work

$$dG = Vdp - SdT \quad (G = \text{Gibbs free energy})$$

Answer the following questions.

15. Fixed mass of an ideal monoatomic gas undergoes given reversible process



For process $A \rightarrow B$ (Use : $\ln(3) = 1.1$)

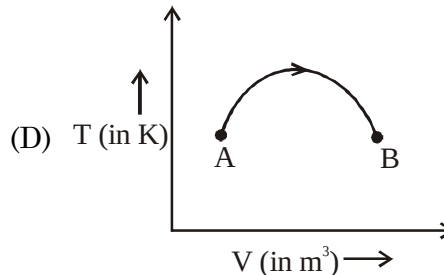
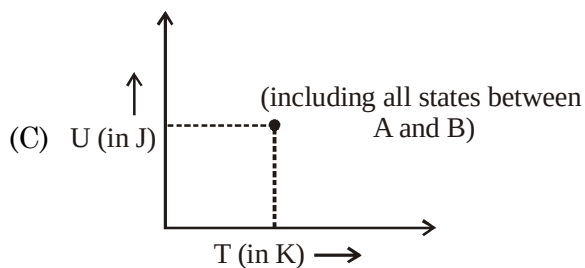
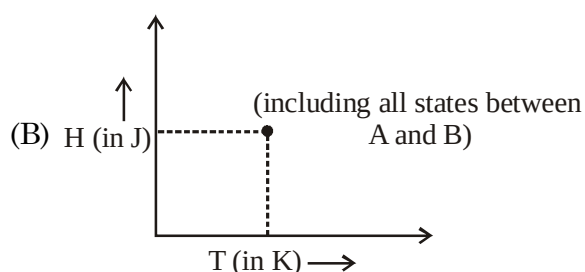
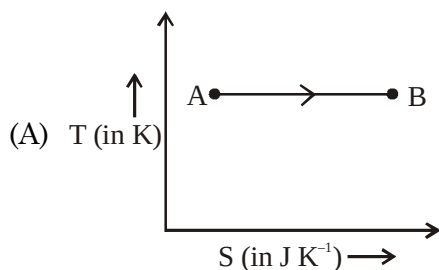
(A) $\Delta G = 0$

(B) $\Delta H = 0$

(C) $\Delta U = 0$

(D) $\Delta G = -13.2 \text{ litre-atm}$

16. For above process $A \rightarrow B$. Select the correct graphs.



Space for Rough Work

SECTION-II : (Maximum Marks: 12)

- This section contains **FOUR** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darken the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

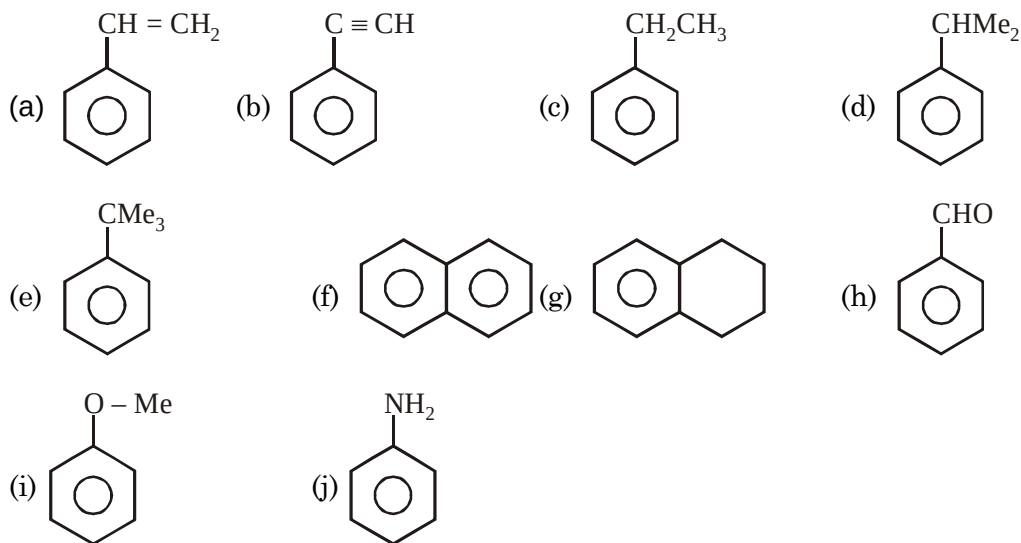
+										-									
●	●	0	0	0	0	0	0	0	0	●	●	●	0	0	●	●	●	●	●
1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. A current of dry air was passed through a series of bulbs containing 1.25 gm of a solute, A_2B (molar mass = 90 gm/mol) in 49 gm of water and then through pure water. The loss in weight of the former series of bulbs was 0.98 gm and in the later series 0.01 gm. The percentage dissociation of solute is.
2. Calculate the minimum volume (in L) of 1M- HNO_3 solution required to completely dissolve 95.25 gm Cu, producing NO. (Cu = 63.5)
3. How many of the following statement is incorrect.
 - (i) B_2H_6 catches fire spontaneously on exposure to air and this property is used in Holme's signal
 - (ii) Alum shows swelling characteristics on heating with Pt-wire loop
 - (iii) $LiH + H_2O \longrightarrow LiOH + H_2$: Proves that LiH is ionic
 - (iv) Li_2SO_4 is insoluble in water
 - (v) $HgCl_2$ gives chromyl chloride test
 - (vi) I_2Cl_6 has same number of atoms in one plane like Al_2Cl_6

Space for Rough Work

4. How many of the following will give benzoic acid with hot acetic KMnO_4 ?



Space for Rough Work

SECTION-IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- The ORS contains a 4×5 matrix whose layout will be similar to the one shown below :

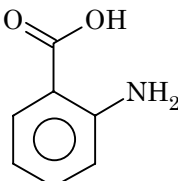
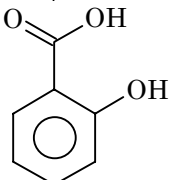
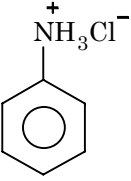
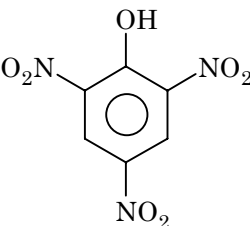
(A)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(B)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(C)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(D)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)

- For each entry in **column-I**, darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the bubble(s) corresponding to all the correct match(es) is (are) darkened

Zero Marks : 0 In all other cases

1.	Column-I	Column-II
(A)		(P) Acid is stronger than benzoic acid.
(B)		(Q) Acid liberates CO_2 when mixed with aqueous NaHCO_3 solution.
(C)		(R) Acid gives a white ppt on mixing with AgNO_3 (aq.)
(D)		(S) Compound soluble in aqueous NaOH .
		(T) Compound is litmus sensitive.

Space for Rough Work

2. Column-I(A) Zn(OH)_2 (B) Fe(OH)_3 (C) Al(OH)_3 (D) Cr(OH)_3 **Column -II**(P) Completely Soluble in NH_4OH

(Q) Soluble in NaOH to form a colourless solution.

(R) Soluble in NaOH in presence of H_2O_2 to form a yellow solution.

(S) Soluble in HCl to form a colourless solution.

(T) Soluble in HCl to form a yellow solution.

Space for Rough Work

PART-3 : MATHEMATICS

SECTION-I(i) : (Maximum Marks : 24)

- This section contains **EIGHT** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :
Full Marks : +3 If only the bubble corresponding to the correct option is darkened.
Zero Marks : 0 If none of the bubbles is darkened.
Negative Marks : -1 In all other cases

1. If $f(x) = (p^2 - 1) [\tan^{-1} x] + 4(q^2 + 2q - 3) \left\{ \frac{1}{2+x^2} \right\} + (p+q) \operatorname{sgn}(x^2 - x + 2)$ is continuous in $\mathbb{R}$ and $f(x_1) = f(x_2) \forall x_1, x_2 \in \mathbb{R}$ then largest value of $|p+q|$ is
 (A) 0 (B) 2 (C) 4 (D) 5
 [Note : $\operatorname{sgn}(y)$, $[y]$ and $\{y\}$ denote signum function, greatest integer function and fractional part function respectively.]
2. Let $\alpha_r = e^{i \frac{2\pi r}{n}}$ ($1 \leq r \leq n$) be the complex number associated with the point A_r on Argand plane, and point B is (2, 0) then the value of $BA_1 \cdot BA_2 \cdot BA_3 \dots BA_n$ is equal to
 (A) n (B) $2^n - 1$ (C) 2n (D) $2n - 1$
3. ${}^{101}C_0 \cdot {}^{100}C_{100} + {}^{101}C_2 \cdot {}^{100}C_{98} + {}^{101}C_4 \cdot {}^{100}C_{96} + \dots + {}^{101}C_{100} \cdot {}^{100}C_0 =$
 (A) ${}^{201}C_{100}$ (B) ${}^{201}C_{100} + {}^{100}C_{50}$ (C) $\frac{{}^{201}C_{100} - {}^{100}C_{50}}{2}$ (D) $\frac{{}^{201}C_{100} + {}^{100}C_{50}}{2}$

Space for Rough Work

4. Consider a point O (6, 2) on a circle $x^2 + y^2 - 4x - 10y + 4 = 0$. Two lines $x - 7y + 8 = 0$ and $x - 2y - 2 = 0$ are drawn from point O, which intersect the given circle at A and B respectively. If slope of line AB is M, then $|M| =$
- (A) $\frac{1}{3}$ (B) $\frac{1}{4}$ (C) $\frac{2}{3}$ (D) 1
5. If $8 \int_0^{\pi/2} \frac{x^2 (\cos x - \sin x) dx}{(\cos x + \sin x)^3} = \pi(k - \pi)$, then k is equal to
- (A) 3 (B) 2 (C) 0 (D) 5
6. In an acute angled triangle ABC, the circle on the altitude AD as diameter cuts AB at P and AC at Q. Then length PQ is equal to (Where Δ & R represents area and circum radius of the ΔABC)
- (A) $\frac{2\Delta}{R}$ (B) $\frac{\Delta}{R}$ (C) $\frac{\Delta}{2R}$ (D) $\frac{\Delta}{4R}$

Space for Rough Work

7. Let $S_r = \frac{r^5 + 3r^4 + 2r^3 + 2}{r(r+1)(r+2)}$ then $\left[\sum_{r=1}^{10} S_r \right] =$

(Where $[.]$ represent Greatest Integer Function)

(A) 386

(B) 385

(C) 388

(D) 390

8. If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar unit vector, each inclined with other at angle $\frac{\pi}{6}$, then volume of tetrahedron whose edge are $\vec{a}, \vec{b}$ and $\vec{c}$ is :

(A) $\frac{\sqrt{3\sqrt{3}-5}}{12}$

(B) $\frac{2\sqrt{3}+5}{12}$

(C) $\frac{3+5\sqrt{2}}{12}$

(D) None

Space for Rough Work

SECTION-I(ii) : (Maximum Marks : 32)

- This section contains **FOUR** paragraphs.
- Based on each paragraph, there are **TWO** questions
- Each question has **FOUR** options (A), (B), (C) and (D) **ONE OR MORE THAN ONE** of these four option(s) is(are) correct.
- For each question, darken the bubble(s) corresponding to all the correct option(s) in the ORS.
- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.
Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks : -1 In all other cases.
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

Paragraph for Questions 9 and 10

Mohan and Sohan plays a series of games and Mohan wins the first game. Probability of a player winning a game is $\frac{2}{3}$ if he wins a previous game. If a player wins 2 consecutive games, then the probability of his winning the next game is $\frac{3}{4}$. Also if a player wins 3 consecutive games, then the probability of his winning the next game is $\frac{4}{5}$. (Assume no game ends in a draw)

9. Find the probability that Mohan wins the fourth game.
- (A) $\frac{2}{5}$ (B) $\frac{173}{270}$ (C) $\frac{17}{27}$ (D) $\frac{79}{135}$
10. If Mohan wins fourth game, find the probability that he has won all the four games.
- (A) $\frac{7}{17}$ (B) $\frac{2}{5}$ (C) $\frac{54}{79}$ (D) $\frac{17}{27}$

Space for Rough Work

Paragraph for Questions 11 and 12

Consider the system of equations :

$$x + \lambda y + 3z = 15$$

$$3x + y - z = 3 + \lambda$$

$$4x + 3y + \lambda z = 17 + \mu$$

11. If the given system is consistent with infinite solution sets, then
 (A) $\lambda = 2$ and $\mu = -3$ (B) $\lambda = 2$ and $\mu = 3$
 (C) $\lambda = -3$ and $\mu = 2$ (D) $\lambda = -3$ and $\mu = -49/2$
12. If the given systems has unique solution set, then possible value(s) of λ will be –
 (A) -2 (B) 3 (C) -1 (D) 4

Paragraph for Questions 13 and 14

L_1 and L_2 are two lines given by the equations $\frac{x-3}{1} = \frac{y-5}{-2} = \frac{z-7}{1}$; $\frac{x+1}{7} = \frac{y+1}{-6} = \frac{z+1}{1}$ respectively then

13. Shortest distance between the lines is
 (A) $\sqrt{29}$ (B) $2\sqrt{29}$ (C) 6 (D) $2\sqrt{31}$
14. The equation of line of shortest distance between the lines is
 (A) $\frac{x-3}{2} = \frac{y-5}{3} = \frac{z-7}{4}$ (B) $\frac{x+3}{2} = \frac{y+5}{3} = \frac{z-7}{4}$
 (C) $\frac{x+3}{2} = \frac{y-5}{3} = \frac{z-7}{4}$ (D) $\frac{x-3}{2} = \frac{y-5}{3} = \frac{z+7}{4}$

Space for Rough Work

Paragraph for Questions 15 and 16

Consider, $f(x) = \begin{cases} x^3 - x, & |x| \geq 1 \\ |x^3 - x|, & |x| < 1 \end{cases}$

Identify which of the following statements(s) is/are correct?

15. Which of the following statement(s) is/are correct?

(A) $f(x^2 - 4x + 5) + f(x^2 + 4x - 5) = 0 \forall x \in \mathbb{R}$

(B) $f(-1 + x^2) = f(1 - x^2) \forall x \in \mathbb{R}$

(C) if the value of $y = p(x + 1)$ cut the graph of $y = f(x)$ at 5 distinct points then complete set of value of p is $\left(0, \frac{1}{4}\right)$

(D) if the value of $y = p(x + 1)$ cut the graph of $y = f(x)$ at 5 distinct points then complete set of value of p is $\left(0, \frac{1}{2\sqrt{3}}\right)$

16. Which of the following statement(s) is/are correct?

(A) $y = f(x)$ is an odd function

(B) $y = f(x)$ is neither even nor odd function

(C) number of extremum of the function $f(x) = 0$ is three.

(D) number of solution of the equation $f(x) - k = 0$ is one where $k \in (-\infty, 0)$

Space for Rough Work

SECTION-II : (Maximum Marks: 12)

- This section contains **FOUR** questions.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct) by darkening the corresponding bubbles in the ORS.

For Example : If answer is -77.25, 5.2 then fill the bubbles as follows.

+									
●	●	0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	5	5	5	●	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	●	●	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

-									
●	●	●	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1	1	1
2	2	2	2	2	●	2	2	2	2
3	3	3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4	4	4
5	5	5	●	5	5	5	5	5	5
6	6	6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9	9	9

- Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If ONLY the correct numerical value is entered as answer.
Zero Marks : 0 In all other cases.

1. If the value of sum $\sum_{K=0}^{1006} \frac{2012-2K}{(K+1) \binom{2013}{K+1}}$ is α , then value of $\alpha + \frac{1}{\binom{2013}{1006}}$, where $\binom{n}{r} = {}^nC_r$

2. Find total number of points of maxima and minima of the function $f(x) = \begin{cases} x^4 - 2x^2 & ; -3 < x \leq 0 \\ 3x^{\frac{2}{3}} - 2x & ; 0 < x < 2 \\ -1 & ; x = 2 \\ |x-3| & ; 2 < x < 4 \end{cases}$

Space for Rough Work

3. Find number of solutions of the equation $3\tan x + x^3 = 2$ in $x \in \left(0, \frac{\pi}{4}\right)$.

4. Consider the system of equations

$$2yz + zx - 5xy = 2$$

$$yz + 2xy - zx = 1$$

$$yz - 2zx + 6xy = 3$$

Find $|xyz|$

Space for Rough Work

SECTION-III : Integer Value Correct Type
No question will be asked in section III

SECTION-IV : (Maximum Marks : 16)

- This section contains **TWO** questions.
- Each question contains two columns, **Column-I** and **Column-II**.
- **Column-I** has **four** entries (A), (B), (C) and (D)
- **Column-II** has **five** entries (P), (Q), (R), (S) and (T)
- Match the entries in **Column-I** with the entries in **column-II**.
- One or more entries in **Column-I** may match with one or more entries in **Column-II**.
- The ORS contains a 4×5 matrix whose layout will be similar to the one shown below :

(A)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(B)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(C)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)
(D)	☐ (P)	☐ (Q)	☐ (R)	☐ (S)	☐ (T)

- For each entry in **column-I**, darken the bubbles of all the matching entries. For example, if entry (A) in **Column-I** matches with entries (Q), (R) and (T), then darken these three bubbles in the ORS. Similarly, for entries (B), (C) and (D).
- For each question, marks will be awarded in one of the following categories :

For each entry in **Column-I**

Full Marks : +2 If only the bubble(s) corresponding to all the correct match(es) is (are) darkened

Zero Marks : 0 In all other cases

1.	Column-I	Column-II
(A)	Area of quadrilateral formed by the tangents and normals drawn at the extremities of latus rectum of parabola $y^2 = 4x$, is equal to	(P) 10
(B)	the angle between the tangent drawn to $(y - 2)^2 = 4(x + 3)$ at the points where it is Intersected by the line $3x - y + 8 = 0$ is $\frac{4\pi}{p}$, then p is divisible by	(Q) 8
(C)	If the line $x - 1 = 0$ is the directrix of the parabola $y^2 - kx + 8 = 0$ then the value of $ k $ is/are	(R) 2
(D)	If P is any point on ellipse $x^2 + 16y^2 = 4$ whose foci are at F_1 and F_2 , then the maximum value of $(PF_1)(PF_2)$ is equal to	(S) 4 (T) 6

Space for Rough Work

2. Let $S = 0$ be the parabola touching x-axis at (1,0) and $y = x$ at (1,1).

Column-I	Column-II
(A) If the slope of axis of the parabola is $\frac{a}{b}$, find $a + b$ (where H.C.F of a and b is one)	(P) 3
(B) If the focus of the parabola is (a, b) then find $5(a + b)$	(Q) 4
(C) If the equation of directrix is $ax + by = 1$, then find $(a - b)$	(R) 1
(D) If foot of directrix of the parabola is (a, b) then find $\frac{25}{2}(a + b)$	(S) 7
	(T) 8

Space for Rough Work